EXPLAINABLE SKIN CANCER DETECTION, SEVERITY PREDICTION AND PERSONALIZED CLINICAL DECISION SUPPORT USING DEEP LEARNING

A PROJECT REPORT

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in partial fulfillment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

Loyola **INSTITUTE OF TECHNOLOGY AND SCIENCE**



ANNA UNIVERSITY: CHENNAI 600 025

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BONAFIDE CERTIFICATE

Certified that this project report **"EXPLAINABLE SKIN CANCER DETECTION, SEVERITY PREDICTION AND PERSONALIZED CLINICAL DECISION SUPPORT USING DEEP LEARNING"** is the bonafide work of **"ANU A (961223104007), MUTHAMIL SELVI S (961223104039), VARSHA N (961223104054), VIJAYA SHALINI M (961223104055)"**

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ABSTRACT

In recent years, the use of deep learning in medical image analysis has become increasingly important for supporting the early identification of skin cancer. This project presents **SkinGuard AI**, a deep learning-based system developed for skin lesion analysis using dermoscopic images. The proposed system combines image quality checking, preprocessing, lesion segmentation, disease classification, explainable artificial intelligence, severity prediction, and personalized clinical decision support in a single workflow. Initially, the input image is checked for basic quality factors such as blur, brightness, contrast, and visibility to ensure that it is suitable for further analysis. The preprocessing is preprocess the image using **BM3D** and **CLAHE**. The lesion region is then extracted using a **U-Net** segmentation model, which helps reduce the influence of surrounding skin and background regions. For disease classification, **EfficientNetB3** is used to classify the lesion into seven categories from the HAM10000 dataset, including melanoma, basal cell carcinoma, actinic keratosis, and other common skin lesions. To make the prediction easier to understand, **Grad-CAM** is applied to highlight the image regions that contribute to the model's decision. Based on the predicted disease and model confidence, the system provides a **Low, Medium, or High severity level** along with personalized clinical guidance. The analysis results are also organized into a PDF report for convenient reference. The proposed system aims to provide an integrated and user-friendly approach to skin lesion assessment while improving the interpretability of deep learning predictions and supporting early medical consultation.

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LIST OF ABBREVATIONS

AI	- Artificial Intelligence
CNN	- Convolutional Neural Network
U-Net	- U-Net Convolutional Network
ISIC	- International Skin Imaging Collaboration
HAM10000	- Human Against Machine with 10000 Training Images
XAI	- Explainable Artificial Intelligence
Grad-CAM	- Gradient-weighted Class Activation Mapping
GPU	- Graphics Processing Unit
CPU	- Central Processing Unit
RGB	- Red, Green and Blue
ReLU	- Rectified Linear Unit
Adam	- Adaptive Moment Estimation
PDF	- Portable Document Format

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

In today's rapidly developing healthcare and digital technology environment, the early detection of skin cancer has become an important challenge. Skin cancer is one of the most common forms of cancer, and identifying suspicious skin lesions at an early stage can help in providing timely medical attention. Dermoscopic images provide detailed information about skin lesions, but examining a large number of images manually can be time-consuming and may lead to variations in interpretation. Therefore, there is a growing need for intelligent systems that can assist in analysing skin lesions accurately and consistently.

Recent advances in deep learning have made it possible to analyse medical images and identify patterns that may not be easily recognized through conventional image examination. However, a prediction alone may not be sufficient in a medical application, as users may also need to understand the region of the image that influenced the model's decision. In addition, providing only a disease label does not give sufficient guidance about the possible severity or the appropriate next step.

To address these issues, **SkinGuard AI** is developed as an integrated skin lesion analysis system using deep learning techniques. The proposed system performs image quality checking and preprocessing using **BM3D** and **CLAHE**, followed by lesion segmentation using **U-Net** and disease classification using **EfficientNetB3**. **Grad-CAM** is incorporated to provide a visual explanation of the classification result by highlighting the important regions of the lesion. Based on the predicted condition, the system also provides a severity level and personalized clinical decision support. The final analysis can be organized into a PDF report, making the results easier to review and share. The overall objective of SkinGuard AI is to support early skin cancer assessment through an automated, explainable, and user-friendly approach while assisting users in making informed decisions about seeking professional medical consultation.

1.2 PROBLEM STATEMENT

Skin cancer is a major health concern, and early identification of suspicious skin lesions is important for timely medical attention. Traditional examination of dermoscopic images mainly depends on manual observation by medical professionals, which can be time-consuming and may vary based on the experience of the examiner. Analysing a large number of skin lesion images manually also makes it difficult to provide quick and consistent results.

Another important issue is that a classification result alone does not provide sufficient information about the affected region or the possible severity of the condition. Users may also find it difficult to understand why a particular disease has been predicted by a computer-based system. In addition, separate tools are often required for image preprocessing, lesion segmentation, disease classification, and result reporting, making the overall analysis less convenient.

Therefore, there is a need for an integrated system that can analyse dermoscopic skin images, identify the lesion region, classify different types of skin lesions, provide a visual explanation of the prediction, estimate the severity level, and offer suitable clinical guidance. The proposed **SkinGuard AI** addresses these requirements by combining image quality checking, **preprocessing with BM3D and CLAHE, U-Net-based segmentation, EfficientNetB3 classification, Grad-CAM-based explanation, severity prediction, personalized clinical decision support, and PDF report generation** within a single workflow. The system is intended to assist users in understanding skin lesion analysis and encourage timely consultation with qualified medical professionals.

Similarly, emotional distress such as anger, fear, or extreme sadness can indicate dangerous situations or mental health crises. However, conventional surveillance systems lack the intelligence to detect these behavioral cues automatically. There is a lack of integrated systems that can simultaneously analyze emotional states and violent actions in real time and trigger immediate alerts. Most existing solutions focus either on facial expression analysis or action recognition separately, without combining both aspects into a unified safety framework. Furthermore, many deep learning models used

in surveillance are computationally heavy, making them unsuitable for real-time applications on low-resource devices.

In addition to these limitations, privacy and scalability issues further complicate the effectiveness of current surveillance systems. Continuous video recording without intelligent filtering leads to large volumes of unnecessary data, making storage and retrieval difficult. This also raises concerns regarding data privacy and misuse of sensitive information. Another major challenge is the lack of adaptability in existing systems. Many surveillance models are trained for specific environments and fail to perform effectively under varying lighting conditions, crowded scenes, or different camera angles. This reduces their reliability in real-world scenarios.

Furthermore, delayed alert mechanisms in traditional systems reduce the chances of timely intervention. Without real-time automated alerts, authorities may not respond quickly to critical situations, increasing the risk of harm. There is also a need for cost-effective solutions that can be deployed widely. High-end surveillance systems require powerful hardware, which increases overall cost and limits accessibility for small-scale or public installations.

1.3 OBJECTIVES OF THE PROJECT

The main objectives of the proposed **SkinGuard AI** system are:

1. **To develop an automated system** for analysing dermoscopic skin lesion images and supporting early skin cancer assessment.
2. **To check the quality of the input image** by analysing factors such as blur, brightness, contrast, noise, and visibility before performing further analysis.
3. **To preprocess the dermoscopic images** by resizing and normalizing them to obtain a suitable input for the deep learning models using BM3D and CLAHE.
4. **To segment the skin lesion accurately** using the U-Net model so that the affected region can be separated from the surrounding skin.
5. **To classify different types of skin lesions** using the EfficientNetB3 deep learning model trained on the HAM10000 dataset.

6. **To provide an understandable explanation of the prediction** using Grad-CAM by highlighting the important regions that influenced the classification result.
7. **To estimate the severity level** of the detected skin condition as Low, Medium, or High based on the analysis results.
8. **To provide personalized clinical decision support** by suggesting appropriate next steps based on the predicted condition and severity.
9. **To generate a structured PDF report** containing the image analysis results, prediction, confidence, severity, explanation, and recommendation.
10. **To integrate all the analysis stages into a single workflow**, making skin lesion assessment easier, faster, and more convenient for users while supporting consultation with qualified medical professionals.

1.4 SCOPE OF THE PROJECT

The scope of the proposed **SkinGuard AI** system is to develop a complete and user-friendly platform for the preliminary analysis of dermoscopic skin lesion images. The system begins with an image quality assessment to determine whether the input image is suitable for further processing by checking factors such as blur, brightness, contrast, noise, resolution, and overall visibility. After quality verification, the image is preprocessed using **BM3D** and **CLAHE** that resizing and normalizing it into a suitable format for deep learning analysis. The system then focuses on identifying the lesion region using a **U-Net-based segmentation model**, which helps separate the affected area from the surrounding skin and improves the quality of subsequent analysis.

The scope further includes the classification of skin lesions using the **EfficientNetB3** model. The system is designed to classify dermoscopic images into seven categories available in the HAM10000 dataset: **Actinic Keratosis, Basal Cell Carcinoma, Benign Keratosis, Dermatofibroma, Melanoma, Melanocytic Nevus, and Vascular Lesion**. Along with the predicted disease, the system provides a confidence value to indicate the model's prediction level. Since understanding a prediction is important in medical image analysis, **Grad-CAM** is incorporated to generate a visual heatmap showing the regions of the lesion that contributed to the classification result.

Another important part of the scope is **severity prediction and personalized clinical**

decision support. Based on the classification result and confidence, the system provides a severity category such as Low, Medium, or High and presents suitable guidance for the user's next step. The system is intended to help users understand the analysis result and recognize when professional medical consultation may be appropriate. It does not attempt to replace a dermatologist or provide a final medical diagnosis.

The project also covers the generation of a **PDF report** containing the important analysis results, including the input image, predicted condition, confidence score, severity level, Grad-CAM explanation, and recommendation. By combining image quality assessment, preprocessing, segmentation, classification, explainability, severity prediction, clinical decision support, and reporting into one workflow, SkinGuard AI aims to provide a convenient preliminary skin lesion assessment system. The project can be further extended in the future with larger and more diverse datasets, improved model architectures, clinical validation, and additional decision-support features.

1.5 EXISTING SYSTEM

In a conventional skin lesion examination process, the patient image or lesion is examined manually by a medical professional. The doctor observes characteristics such as color, shape, border and appearance of the lesion.

Digital dermoscopy and image-based systems can assist this process, but many systems concentrate on only one stage, such as classification.

A basic computer vision system may directly send the complete image to a classifier. In such a setup, unwanted background information may also be considered by the model.

Another limitation is explainable. If the model predicts a particular class, the user may not know which part of the image influenced that prediction.

1.6 LIMITATIONS OF EXISTING SYSTEM

- Manual examination requires expert involvement.

- Image quality can affect visual analysis.
- Background regions may influence automated classification.
- Direct classification may not clearly isolate the lesion.
- Prediction results may be difficult for non-technical users to understand.
- Some systems do not provide visual explanation of predictions.

These limitations motivated the development of SkinGuard AI.

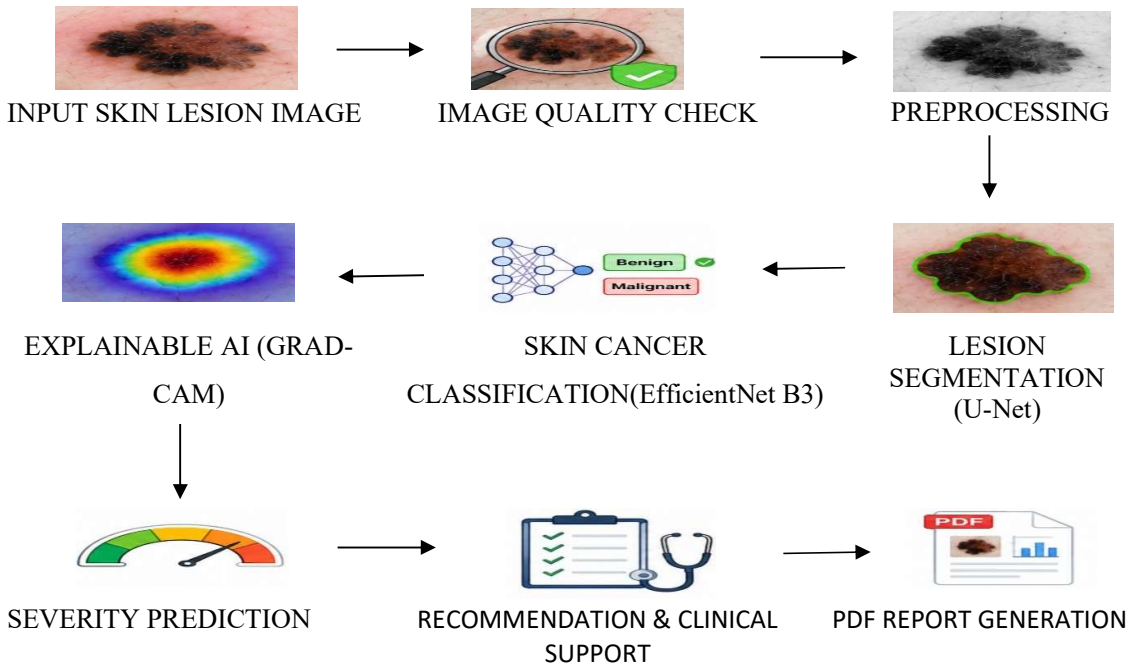
1.7 PROPOSED SYSTEM

The proposed system combines different image processing and deep learning techniques into a single workflow. The input skin image is first checked for quality.

After preprocessing, the lesion region is extracted using a U-Net segmentation model. The resulting image is then passed to the EfficientNetB3 classification model.

The classification model predicts the lesion category and provides a confidence value.

The overall workflow can be represented as:



1.8 ADVANTAGES OF PROPOSED SYSTEM

The proposed system provides the following advantages:

- Provides a structured image analysis workflow.
- Automatically separates the lesion region.
- Uses a pretrained EfficientNetB3 architecture for classification.
- Uses HAM10000 images for model training.
- Provides classification confidence.
- Provides visual explanation through Grad-CAM.
- Combines multiple modules in one application.
- Provides a simple user interface.
- Can be extended with additional models and features.

1.9 APPLICATIONS

The system can be used for:

- Academic demonstration of medical image analysis.
- Deep learning project demonstrations.
- Computer vision research.
- Skin lesion image analysis experiments.
- Explainable AI demonstrations.
- Educational purposes for understanding image segmentation and classification.

1.10 ORGANIZATION OF THE REPORT

Chapter 1 – Introduction

This chapter explains the project background, problem statement, objectives, scope and the proposed solution.

Chapter 2 – Literature Survey

This chapter discusses existing research and techniques related to skin lesion analysis, segmentation, classification and explainable AI.

Chapter 3 – System Analysis and Methodology

This chapter explains the existing system, the proposed system, architecture and the complete working methodology.

Chapter 4 – Hardware and Software Requirements

This chapter describes the hardware and software tools used for developing the project.

Chapter 5 – Implementation and Results

This chapter explains the implementation of each module and presents the obtained results and screenshots.

Chapter 6 – Conclusion and Future Scope

This chapter summarizes the completed work. Discusses possible improvements.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

In recent years, deep learning-based medical image analysis has gained significant attention in the field of dermatology. Skin cancer is one of the major health concerns worldwide, and early identification of suspicious skin lesions is important for timely medical consultation and treatment. Dermoscopic imaging provides detailed visual information about skin lesions and is widely used for examining different types of skin conditions. However, the manual analysis of dermoscopic images requires considerable time and experience, and differences in visual interpretation may affect the consistency of the assessment.

The focus of this chapter is to review existing research related to skin lesion segmentation, skin cancer classification, deep learning-based medical image analysis, and explainable artificial intelligence. The review covers different segmentation techniques, classification models, datasets, explainability methods, and their limitations. This analysis helps to identify the research gaps and provides the basis for the proposed **SkinGuard AI** system, which combines U-Net for lesion segmentation, EfficientNetB3 for skin lesion classification, Grad-CAM for visual explanation, and severity-based clinical decision support.

Traditional image processing techniques have been used for identifying lesion boundaries and extracting visual features from dermoscopic images. However, these methods may be affected by variations in illumination, skin colour, lesion shape, hair, and image quality. With the development of deep learning, convolutional neural networks have provided improved performance by learning useful features directly from medical images. Segmentation networks such as U-Net have become widely used for extracting lesion regions, while transfer learning models have been applied for classifying different skin lesion categories.

Another important area of recent research is explainability. Although deep learning models can achieve good classification performance, their predictions may be difficult for users to understand because the decision-making process is not directly visible. Techniques such as Grad-CAM provide visual representations of the regions that contribute to a model's prediction. Therefore, an integrated system that performs image quality assessment, segmentation, classification, explanation, severity prediction, and clinical guidance can provide a more complete approach to preliminary skin lesion analysis.

2.2 SKIN LESION IMAGE ANALYSIS

Skin lesion analysis involves examining an image and identifying the region that represents the lesion. Important visual characteristics may include:

- Colour
- Shape
- Texture
- Border
- Size
- Internal patterns

However, these characteristics can vary significantly between images. Lighting, camera quality and background can also affect the appearance.

Because of these variations, preprocessing and segmentation can be useful before classification.

2.3 IMAGE SEGMENTATION

Image segmentation is the process of separating an important region from the rest of an image.

In skin lesion analysis, segmentation is used to identify the lesion region from surrounding skin.

Traditional methods include:

- Thresholding
- Region Growing
- Edge detection
- Clustering

Deep learning based segmentation models have become popular because they can learn complex patterns from training images.

2.4 U-NET

U-Net is a convolutional neural network architecture commonly used for image segmentation.

It contains two main parts:

- Encoder
- Decoder

The encoder extracts features from the input image, while the decoder gradually reconstructs the spatial information required to produce the segmentation mask.

Skip connections connect corresponding encoder and decoder layers. These connections help preserve useful spatial information.

In SkinGuard AI, U-Net is used to identify the lesion region before classification.

2.5 SKIN LESION CLASSIFICATION

Once we have completed segmentation, we move to classification. Classification algorithms learn the pattern from labeled images and classify an unlabeled image.

For the current project, we use the HAM10000 dataset which consists of images that belong to different types of skin lesions.We use EfficientNetB3 for classification**.**

2.6 EFFICIENTNETB3

It is a family of convolutional neural networks designed to achieve a balance between model size and performance.

EfficientNetB3 is used in this project because it provides a relatively efficient architecture while still being capable of learning detailed image features.

The pretrained ImageNet weights are used as the starting point. The original classification head is replaced with a custom layer that produces seven output classes corresponding to the classification dataset.

2.7 HAM10000 DATASET

Dataset HAM10000 is publicly available dermatoscopic images of commonly observed pigmented skin lesions.

mages are preprocessed to resize them to 224 × 224 pixels and normalize them for input into the classification model.

This data is split into:

- Training set: 8,012 images
- Validation set: 1,001 images
- Testing set: 1,002 images

Splitting is done using stratification.

2.8 EXPLAINABLE AI

Deep learning models can produce accurate predictions, but understanding the reason behind a prediction can be difficult.

Explainable AI methods attempt to make model decisions easier to interpret.

In this project, Grad-CAM is used for this purpose.

2.9 GRAD-CAM

This technique generates a heatmap showing what parts of the image led to the prediction of the chosen class.

Heatmap may be superimposed on the original image so that the user could see which region of the picture impacted the network's decision.

In SkinGuard AI, Grad-CAM is applied after classification.

2.10 LITERATURE GAP

From the study of existing approaches, it can be observed that different techniques are often developed for individual tasks.

Some approaches concentrate mainly on segmentation, while others focus on classification. Similarly, explainability may be treated separately.

SkinGuard AI attempts to connect these stages into a single workflow:

Segmentation + Classification + Explainability

2.11 Comparative Analysis of Existing Studies

Lots of researchers have dug into using deep learning for automatic skin lesion analysis. Most folks seem to stick with classification—basically, they want to teach machines to tell what kind of skin issue is in an image. Others spend their energy on cutting out the lesion (segmentation) or making the AI's decisions more understandable. The SkinGuard AI system aims to roll all these pieces together, so you don't have to pick just one.

Traditional machine learning works with dermoscopic image datasets and relies on feature extraction for disease classification. But there's a catch—someone has

to manually decide which features matter, and the models don't always generalize well.

Then you've got the classic CNN-based approaches. They're pretty good at automated disease classification but struggle to explain their decisions. It's like getting the answer but not knowing why.

Transfer learning steps things up a bit. Researchers use datasets like HAM10000 and pretrained CNNs to boost classification performance. The drawback? It's still mainly about classification—they don't go the whole nine yards when it comes to analyzing everything.

On the segmentation side, U-Net models do a solid job of finding and marking the lesion regions in dermoscopic images. But just segmenting doesn't help you actually diagnose anything—it stops short.

EfficientNet-based models, especially on datasets like HAM10000, handle multi-class skin lesion classification. Yet, they don't always show you where the lesion is or why they picked their answers.

Grad-CAM comes in as a tool for visual explanations, showing you which parts of the image the model paid attention to. It's neat, but it usually works as an add-on rather than being part of a full diagnostic setup.

SkinGuard AI takes ideas from all these approaches. It uses HAM10000, U-Net, EfficientNetB3, and Grad-CAM, bundling segmentation, classification, explainability, severity analysis, and recommendations into one package. The goal is to help doctors—not replace them—with decision support.

When you look at past studies, it's clear deep learning works for skin lesion analysis. CNNs and transfer learning are great for making quick classifications, while U-Net really nails locating the lesion. But most systems only tackle part of

the problem. SkinGuard AI fills that gap—combining segmentation, classification, visual explanation, and extra information into one web-based tool. It gives you a predicted category, shows why, and adds supportive details. Compared to systems that only classify or segment, SkinGuard AI serves up a more rounded, practical solution focused on helping users get the full picture.

2.12 Summary of Literature Survey

AI and deep learning are really making waves in automated skin lesion analysis and skin cancer detection. Researchers have thrown a lot at this—everything from image preprocessing to segmentation, classification, and even explainable AI. It's all about pushing computer-aided diagnosis systems to be more accurate and trustworthy.

Now, traditional image processing still has its place. It can boost image quality and help spot lesion regions, but it's not perfect. Things like different skin tones, weird lighting, and odd-shaped lesions can trip it up. That's where deep learning, especially U-Net, steps up. U-Net nails segmentation, carving out the lesion from the rest of the skin with much better accuracy.

When it comes to figuring out what kind of skin lesion you have, CNNs and transfer learning models like EfficientNet really shine. Building a powerful CNN from scratch takes a ton of data and time, but with transfer learning, you cut out a lot of that heavy lifting and still get great results.

The survey also drives home how important it is to explain what these AI systems are actually doing, especially in medicine. Doctors and users don't want black boxes. Tools like Grad-CAM help by showing which parts of an image influenced the AI's decision, making everything more transparent and easier to trust.

So, putting all that together, the SkinGuard AI system brings these elements under one roof: image preprocessing, U-Net for segmentation, EfficientNetB3 for classification, and Grad-CAM for visualization. The goal is simple—let the system handle automated lesion classification, but make sure the process stays clear and understandable for everyone involved.

CHAPTER 3

SYSTEM ANALYSIS AND METHODOLOGY

3.1 INTRODUCTION

This chapter presents the system analysis and methodology followed for the development of the proposed SkinGuard AI system. System analysis is an important stage in software development as it helps to understand the requirements of the proposed system, identify the limitations of the existing approach, and determine the necessary components for developing an effective solution. In this project, the analysis focuses on the requirements involved in processing dermoscopic skin images and presenting the obtained results in a structured manner.

The proposed system consists of a sequence of processing stages through which the uploaded dermoscopic image is analyzed. The methodology defines how the input image is handled from the initial upload to the generation of the final analysis report. The major stages include image quality checking, image preprocessing, lesion segmentation, disease classification, visual explanation using Grad-CAM, severity assessment and report generation. Each stage performs a specific task and provides the required information for the subsequent stage.

The methodology also describes the techniques and components used at each stage of the system. Image preprocessing is performed to bring the input image into a suitable format for further processing. Segmentation is used to identify the relevant lesion region, while the classification stage determines the disease category based on the features learned from the training data. The classification output includes the predicted disease and its confidence value. The system further uses Grad-CAM to provide a visual representation of the regions that contribute to the classification result.

System analysis also considers the functional requirements of the application, including image uploading, processing, classification, result visualization and report generation.

The system is organized into separate modules so that each processing stage can be developed, tested and maintained independently. This modular approach also helps to maintain a clear flow of information between the different components of the application.

The methodology followed in this project is designed to provide a systematic flow from input image to final output. The uploaded image is first examined and prepared for processing, followed by segmentation and classification. The generated results are then combined with the visual explanation and other analysis information before being presented to the user through the application and generated report. This chapter explains the analysis of the existing and proposed systems, system requirements, system design, workflow and the methodology used for implementing the different stages of SkinGuard AI.

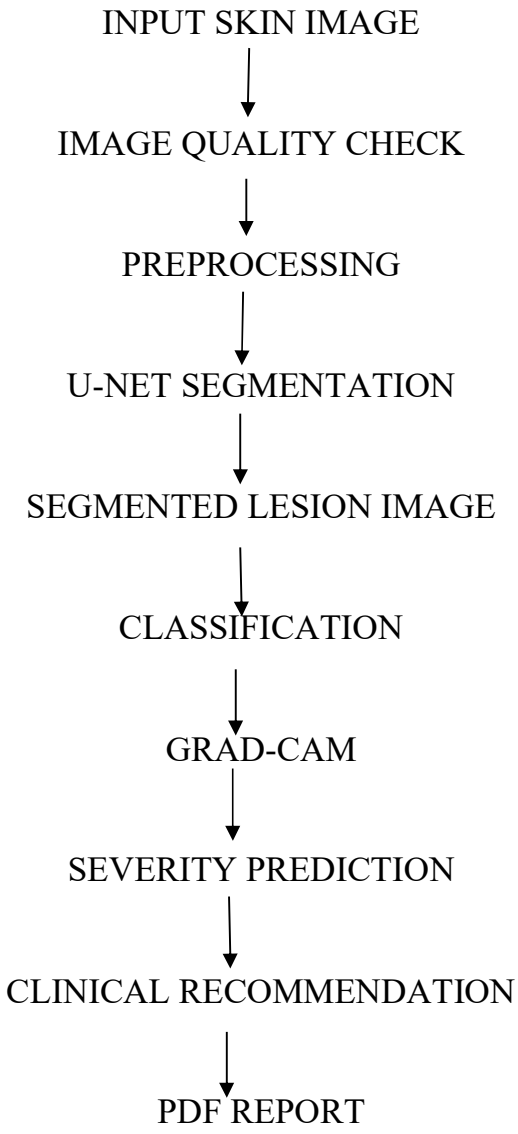
Overall, this chapter provides the technical foundation for understanding how the proposed system is structured and how the different modules work together to process dermoscopic images and generate the final analysis results.

3.2 SYSTEM ARCHITECTURE

The system is made up of the following modules:

1. User Input
2. Image Quality Analysis
3. Image Preprocessing
4. Lesion Segmentation
5. Lesion Classification
6. Grad-CAM Explanation
7. Severity Prediction
8. Recommendation and Clinical Support
9. Pdf Report

SYSTEM FLOW



3.3 IMAGE QUALITY MODULE

The modules that make up the system include:

Some of the factors that the quality step can take into account are:

- Image size
- Brightness
- Blur
- Contrast

- Basic image validity

If the image is found to be appropriate, it proceeds to the next step.

3.4 PREPROCESSING MODULE

The input image is prepared before being passed to the deep learning models. This preprocessing is worked based on BM3D and CLAHE.

The main operations include:

- Reading the image.
- Converting the colour format.
- Resizing the image.
- Normalizing pixel values.

Preparing the image as a model input. For classification, the image is resized to 224 × 224 × 3.

For segmentation, the image is resized according to the U-Net input dimensions.

3.5 SEGMENTATION USING U-NET

The preprocessed image is given to the trained U-Net model. The model produces to an prediction mask. A threshold is applied to convert the prediction into a binary mask.

The mask represents the detected lesion region. The mask is then resized back to the original image dimensions and overlaid on the image. This makes the lesion region easier to visualize.

3.6 CLASSIFICATION USING EFFICIENTNETB3

After segmentation, the image is given to the EfficientNetB3 classification model.

The model contains a pretrained EfficientNetB3 base network followed by:

- Global Average Pooling
- Dropout
- Dense layer
- Dropout
- Seven-class Softmax output layer

The model predicts probabilities for the seven classes.

The class having the highest probability is selected as the predicted class.

3.7 MODEL TRAINING

The HAM10000 dataset is divided into training, validation and testing data.

The classification model is trained using:

Optimizer: Adam
Loss Function: Sparse Categorical Crossentropy
Architecture: EfficientNetB3
Output: Seven classes

During training, validation performance is monitored.

The best model is saved using a model checkpoint.

Early stopping is also used to prevent unnecessary training when the validation performance stops improving.

3.8 CLASSIFICATION OUTPUT

Here's what you'll see from the classification module:

- The class index

- The name of the disease or class
- The model's confidence as a percentage

So, if the model detects Melanoma, it might show something like this:

Predicted Class: Melanoma

Confidence:82.35%

Your results will always depend on your particular image and how the model learned from its data.

3.9 GRAD-CAM MODULE

Once you get a prediction, the Grad-CAM module steps in to explain things. It creates a heatmap that highlights which parts of your image influenced the model's decision.

This heatmap sits right on top of your original image, showing you exactly where the model was paying attention. Basically, you get to see what the model was "looking at" when it made its call.

3.10 RULE-BASED INTERPRETATION

A simple rule-based module can be placed after the model output.

For example, the system can consider the predicted class and confidence value and generate an understandable message.

The rule-based module does not train a separate model.

It simply applies predefined conditions to the model output.

Example:

```
IF confidence >= threshold
   Display prediction with higher confidence message
ELSE
```

Display result with caution message

This part is mainly used to make the output easier to understand.

3.11 FLASK WEB APPLICATION

Flask is used to connect the machine learning models with the web interface.

The application contains separate routes for the different stages.

The user can upload an image and move through the analysis process.

The main application modules include:

- Home page
- Image quality page
- Preprocessing page
- Segmentation page
- Classification page
- Grad-CAM page

This modular structure makes the project easier to maintain.

3.12 SYSTEM WORKFLOW

SkinGuard AI runs through a step-by-step process to analyze dermoscopic skin lesion images. It starts with a simple upload and guides each image through several stages, beginning with a quality check and finishing with a clear recommendation and report.

First, users upload a dermoscopic image through the web app. The system immediately checks the image's quality, looking at things like resolution, sharpness, brightness, contrast, and noise. If the image doesn't pass this first scan, it won't go any further, so nothing gets analyzed unless it's clear enough.

Next, the system preprocesses the image. It resizes it to 224 by 224 pixels and puts it in a format the model can work with. At this point, the pixel values get normalized so the deep learning models receive consistent input every time.

Now, the U-Net segmentation model steps in. It identifies the lesion's location, draws out a segmentation mask, and clearly separates the lesion from the surrounding skin—focusing the next steps on where it matters.

After segmentation, both the processed image and the lesion region go to the EfficientNet-B3 classification model. This part predicts the type of skin lesion, picking from seven classes. It also gives a probability score for each prediction—so you know how confident the model is.

Once the system has a prediction, it uses Grad-CAM to create a heatmap—basically, a visual explanation showing which parts of the image influenced the decision. This makes the whole process more transparent and easier to interpret.

With the classification in hand, the rule-based interpretation module determines how severe the lesion is by applying a set of predefined criteria. The system assigns a severity level and offers a recommendation about what to do next. All these results show up in the web application, and users can generate a PDF report for their records.

3.13 FUNCTIONAL REQUIREMENT

Functional requirements spell out what SkinGuard AI needs to do to actually work. Here's how it breaks down:

1. Image Upload

Users can upload a dermoscopic skin lesion image through the web app. That image kickstarts all the analysis steps.

2. Image Quality Assessment

SkinGuard AI checks the uploaded image for things like resolution, sharpness, brightness, contrast, and noise. If the image isn't good enough, you'll know right away—it won't move forward.

3. Image Preprocessing

Once the image is in, SkinGuard AI resizes it and normalizes the pixel values so it's ready for the deep learning models. This is worked based on BM3D and CLAHE.

4. Skin Lesion Segmentation

A U-Net model zeroes in on the lesion area, separating it from the rest of the skin.

5. Skin Lesion Classification

The EfficientNet-B3 model takes over next, classifying the lesion image into one of the known skin lesion categories.

6. Probability Generation

The system shows predicted probability values for each classification choice—so you can see how confident the model is.

7. Explainable AI Analysis

SkinGuard AI creates a Grad-CAM visualization that highlights which parts of the image influenced the model's prediction.

8. Severity Interpretation

Based on set rules and the classification result, the system determines how severe the lesion is.

9. Recommendation

SkinGuard AI offers a recommendation rooted in the predicted class and severity. It's just guidance—don't take it as a full medical diagnosis.

10. Result Display

Everything from quality check, segmentation, classification, probability values, Grad-CAM visualization, severity, and recommendations are shown in the web app.

11. PDF Report Generation

For documentation and future reference, SkinGuard AI generates a structured PDF report with all the key analysis outcomes.

3.14 DATA FLOW OF THE PROPOSED SYSTEM

SkinGuard AI handles an image from start to finish. It all starts when someone uploads a dermoscopic skin lesion photo through the Flask web app. First thing, the system checks the picture's quality—looking for basic stuff like clarity.

If the image is good enough, the system moves it to the next step: preprocessing. This part transforms the picture into the right format, shrinks it down to 224 × 224 pixels, and gets the pixel values ready.

With preprocessing done, the image heads to the U-Net segmentation module. U-Net creates a mask, highlighting exactly where the lesion sits. That mask lets the system zero in on the area that matters most.

Now, the focused image goes to the EfficientNet-B3 classification module. This model predicts the lesion type and gives you confidence scores for each possible category.

Then the classification gets visual backup from the Grad-CAM module, which cooks up a heatmap showing which parts of the image steered the prediction.

After all that, the Rule-Based Interpretation module kicks in. It reads the classification and heatmap, checks against its set of rules, decides how severe the lesion is, and writes up a recommendation.

In the end, the results swing back to the Flask web app. The user sees the entire analysis on the screen, and if they want, grabs a PDF report with all the important details.

3.15 MODULE INTERACTION

SkinGuard AI isn't just one big block of code—it's a collection of smaller pieces that work together, each doing its own job and passing the results to the next. This setup keeps things organized and makes it way easier to work on and fix parts of the system.

Here's how everything comes together: First, the Image Upload Module greets you. You hand it your dermoscopic image, and it gets things started by sending the picture off for a quality check. Next, the Image Quality Assessment Module takes a look, decides if the image measures up, and weeds out anything that doesn't meet the mark before it goes any further.

If your image gets the green light, the Preprocessing Module steps in. It does some basic housekeeping: resizing, converting formats, normalizing pixels—just making sure the image is ready for the main analysis.

After that comes the U-Net Segmentation Module. This one zooms in on the lesion and creates a mask, highlighting just the area that matters. With the region of interest clearly marked, the EfficientNet-B3 Classification Module analyzes

the image and predicts what kind of skin lesion it's looking at. It doesn't just blurt out an answer—it gives probabilities for each category.

But SkinGuard AI doesn't leave you guessing why a prediction was made. The Grad-CAM Module steps in to show which areas of the image influenced the decision, offering a visual breakdown of the model's thought process.

Once the system has the classification, the Rule-Based Interpretation Module takes over. It applies a set of established rules to figure out how serious the detected lesion is and recommends what to do next.

All of this happens behind the scenes, but you see it through the Flask Web Application Module, which ties everything together and presents the results clearly on the user interface. When the process wraps up, the PDF Report Module gathers the findings and builds a structured report, so you have a detailed summary of everything in one spot.

3.16 SUMMARY

Here's a quick wrap-up for this chapter: I walked you through how SkinGuard AI works—from the moment a dermoscopic image is uploaded, all the way through checking image quality, preprocessing the image with BM3D and CLAHE, segmenting with U-Net, classifying with EfficientNet-B3, adding Grad-CAM for explainability, interpreting severity, and offering recommendations.

I broke down what the system needs to do, like its main functions, and also covered those extra qualities like usability, speed, reliability, maintainability, interpretability, and scalability. I laid out the hardware and software requirements you'll need to get SkinGuard AI up and running.

CHAPTER 4

HARDWARE AND SOFTWARE DESCRIPTION

4.1 INTRODUCTION

SkinGuard AI brings together the right hardware and software to handle everything from segmenting skin lesions and classifying diseases to explainable AI analysis, predicting severity, and supporting clinical decisions. For lesion segmentation, it relies on U-Net, while EfficientNetB3 handles disease classification. Python and TensorFlow power the model development and execution behind the scenes. The web application runs on Flask, making it easy for users to interact with the system.

Every hardware and software choice is made to ensure the system processes images smoothly, trains models effectively, and delivers fast, clear predictions and visual results.

4.2 HARDWARE DESCRIPTION

Here's what you need to run SkinGuard AI – the Explainable Skin Cancer Detection System. The system isn't too demanding, but it does need a computer with enough muscle to handle image processing and deep learning smoothly.

Hardware Requirements:

- **Processor:** Intel Core i5 or better
- **RAM:** At least 8 GB, but you're better off with 16 GB or more
- **Storage:** 256 GB minimum (SSD preferred)
- **GPU:** An NVIDIA GPU really speeds up deep learning work
- **Monitor:** An HD display does the job
- **Camera:** Any good webcam or digital camera for capturing skin images
- **Keyboard:** Standard keyboard
- **Mouse:** Standard mouse.

4.3 HARDWARE SYSTEM REQUIREMENTS

Here's what you need to run the SkinGuard AI – Explainable Skin.

Cancer Detection System:

- **Processor:** Intel Core i5 or better
- **RAM:** At least 8 GB (16 GB or more is even better)
- **Storage:** 256 GB minimum, SSD preferred
- **Graphics Card:** NVIDIA GPU is recommended, especially if you want faster deep learning training
- **Monitor:** 14-inch HD display, or bigger
- **Camera:** You'll need a webcam or a digital camera to capture skin images
- **Keyboard:** Any standard keyboard works
- **Mouse:** Any standard mouse
- **Internet Connection**: Needed for downloading datasets and software packages.

4.4 SOFTWARE DESCRIPTION

SkinGuard AI — the Explainable Skin Cancer Detection System — runs on a mix of Python and web development tools. Here's what's under the hood: the software handles everything from cleaning and prepping images, to cutting out lesions, classifying diseases, showing explainable results, and running the web interface.

Here's what you need to get it working:

- **Operating System:** Windows 10 or 11
- **Programming Language:** Python
- **Web Framework**: Flask
- **Deep Learning:** TensorFlow and Keras
- **Segmentation Model:** U-Net
- **Classification Model:** EfficientNetB3
- **Image Processing:** BM3D, CLAHE
- **Data Handling:** NumPy and Pandas

- **Machine Learning:** Scikit-learn
- **Frontend:** HTML, CSS, and JavaScript
- **Development Environment:** Visual Studio Code
- **Dataset:** HAM10000
- **Web Browser:** Google Chrome or Microsoft Edge.

4.4.1 PYTHON

Python drives everything behind the SkinGuard AI – Explainable Skin Cancer Detection System. From image preprocessing and managing datasets, to training deep learning models and generating predictions, Python handles it all. The team used TensorFlow and Keras for building both the U-Net segmentation and EfficientNetB3 classification models. For image processing, they leaned on OpenCV, while NumPy, Pandas, and Scikit-learn took care of number crunching, organizing datasets, and splitting the data. On top of that, they used Python with Flask to build out the web interface, making everything accessible from a browser.

Here's what Python actually does in this project: it resizes and preprocesses images, processes the HAM10000 dataset, runs U-Net for segmenting skin lesions, and uses EfficientNetB3 to classify diseases. Python also handles the training and evaluation of these models, powers Grad-CAM visualizations for explainability, and runs the rule-based analysis. Through Flask, it connects the AI models with the frontend, tying the whole system together.

4.4.2 FLASK

Flask is a simple Python web framework, and it's what we used to build the web app for SkinGuard AI—our explainable skin cancer detection system. Basically, Flask connects the frontend to our machine learning models.So, in this setup, Flask takes care of user requests, image uploads, moving between pages, and talking to both the U-Net segmentation and EfficientNetB3 classification models. It also sends prediction results and processed images back to the user, so everything flows smoothly.

4.4.3 TENSORFLOW AND KERAS

TensorFlow and Keras are the big players when it comes to deep learning in the SkinGuard AI — Explainable Skin Cancer Detection System. They give you everything you need to put together, train, test, and launch the models behind the system.

Here, TensorFlow and Keras come in handy for building the U-Net model, which handles skin lesion segmentation, and the EfficientNetB3 model, which takes care of skin lesion classification. These frameworks also make it easier to optimize models, check how well they're doing, make predictions, and save trained models so you can use them again later.

4.4.4 OPENCV

OpenCV, short for Open Source Computer Vision Library, handles a lot of the image processing work in the SkinGuard AI system. With OpenCV, you can read, resize, convert, and prep skin images before feeding them into deep learning models.In this project, OpenCV steps in during image preprocessing, segmentation, and prediction. Basically, it gets the images into the right format and size so models like U-Net and EfficientNetB3 can use them without any trouble.

4.4.5 NUMPY AND PANDAS

NumPy and Pandas play a big role in the SkinGuard AI system. Both are Python libraries that help with crunching numbers and managing data. Before any images or labels go to the deep learning models, these tools get everything cleaned up and ready.

NumPy handles the heavy lifting with image data. It stores images as arrays, handles normalization, reshapes data, and takes care of any math operations needed for preprocessing or predictions.

Pandas deals with anything organized in tables—think spreadsheets or CSV files. In this project, it reads the HAM10000 metadata, pulls out image IDs and disease labels, and keeps the classification dataset neat and accessible.

4.4.6 SCIKIT-LEARIN

Scikit-learn is a Python library that makes life easier when you're working with machine learning tasks like data preprocessing, splitting datasets, and handling labels—in this case, for the SkinGuard AI system. Before training the deep learning models, Scikit-learn steps in to help get the data organized.

In this project, the library's main job is to split the HAM10000 dataset into training, validation, and testing sets. It also handles label encoding, which just means it turns the disease names into numbers, so the classification model understands what it's looking at.

4.5 SOFTWARE ARCHITECTURE AND WORKFLOW

SkinGuard AI — Explainable Skin Cancer Detection System — is built on a modular design. The system pulls together image processing, deep learning, explainable AI, and runs everything through a web interface. It runs on Python and Flask, while TensorFlow and Keras handle the heavy lifting for deep learning tasks.

Software Architecture

User Interface

You interact with the system using a web page built from HTML, CSS, and JavaScript. You can upload or take a photo of your skin, then see each step—preprocessing, segmentation, classification, and analysis—unfold right on screen.

Flask Application Layer

This is the backend. It manages everything behind the scenes: moving between pages, sending images to the right modules, and making sure the frontend talks properly to the AI.

Image Quality and Preprocessing Module

First up: this module checks if the photo is good enough to process. If it passes, the module resizes, normalizes, and tweaks the image to get it ready for analysis.

Segmentation Module

Next, the U-Net deep learning model kicks in, identifying and separating the skin lesion from the rest of the image.

Classification Module

After segmentation, EfficientNetB3 classifies the lesion, pinning it to a specific disease category.

Explainable AI Module

Nobody likes a black box, so this module uses Grad-CAM to generate a heatmap. This helps you see which parts of the photo influenced the classification.

Severity and Recommendation Module

The system takes the classification and its confidence, follows preset rules, and offers detailed info on severity as well as recommendations tailored to what it found.

Workflow

User

↓

Image Upload / Capture

↓

Image Quality Check

↓

Image Preprocessing

↓

U-Net Segmentation

↓

Lesion Region Extraction

↓

EfficientNetB3 Classification

↓

Disease Prediction + Confidence

↓

Grad-CAM Explainability

↓

Severity Prediction

↓

Rule-Based Recommendation

↓

Final Result / Report

4.6 SYSTEM IMPLEMENTATION DETAILS

SkinGuard AI – Explainable Skin Cancer Detection System runs as a web app built with Python, Flask, TensorFlow/Keras, OpenCV, HTML, CSS, and JavaScript. Everything's divided into modules, so it stays organized and easier to update over time.

1. User Interface

The front end uses HTML5, CSS3, and JavaScript. Users see pages for uploading images, checking quality, preprocessing, segmentation, classification, Grad-CAM visualization, severity analysis, and getting recommendations.

2. Image Preprocessing

Once someone uploads a skin image, BM3D and CLAHE handle the processing. The app resizes the image to fit what the model expects, normalizes it, and then pushes it to the deep learning modules.

3. Lesion Segmentation

For segmentation, the system uses a U-Net model through TensorFlow/Keras. This model finds the lesion area and produces a mask. The processed mask shows where the lesion is and displays right on the web page.

4. Skin Lesion Classification

EfficientNetB3 takes over for classification. The trained model looks at the processed lesion image and predicts the specific skin lesion category. You also get a confidence score with the prediction.

5. Explainable AI

Grad-CAM offers clarity on why the model chose its answer. It generates a heatmap, highlighting the parts of the image that influenced the prediction the most.

6. Rule-Based Analysis

There's a rule-based module that takes the classification output and other prediction data, then delivers severity info and recommendations. These rules come built into the app—they aren't trained separately.

7. Flask Backend

Flask keeps backend tasks running and ties together the frontend and AI parts. Different Flask routes manage each stage, making sure processed results move smoothly from page to page.

8. Model Integration

The trained U-Net and EfficientNetB3 models are stored as files. Flask loads them during prediction, so there's no need to retrain every time—just use the models right away.

9. Result Generation

Finally, users see the segmented image, predicted lesion type, confidence score, Grad-CAM visualization, severity details, and personalized recommendations—all laid out on the web interface for easy interpretation.

4.7 SYSTEM REQUIREMENT

To run the SkinGuard AI – Explainable Skin Cancer Detection System, you need the right hardware and software to handle image processing, deep learning model tasks, predictions, and everything that happens through the web interface.

Hardware Requirements

Component	Minimum
Processor	Intel Core i5 or similar
RAM	8 GB
Storage	256 GB

GPU	Not required for predictions
Display	HD Display
Camera	Basic webcam
Internet	Needed to set up

Software Requirements

Software	**Requirement**
Operating System	Windows 10/11 or Linux
Programming Language	Python 3.10 or newer
Web Framework	Flask
Deep Learning	TensorFlow or Keras
Image Processing	BM3D, CLAHE
Data Processing	NumPy, Pandas
Machine Learning	Scikit-learn
Frontend	HTML5, CSS3, JavaScript
IDE	Visual Studio Code
Web Browser	Google Chrome or Microsoft Edge
Dataset	HAM10000

Functional Requirements

This system does a lot more than just accepting skin images from users. Here's what it covers:

- Checks each image for quality, and preprocesses it.
- Finds the lesion using a U-Net segmentation model.

- Classifies the lesion with EfficientNetB3.
- Shows the predicted disease type and confidence score.
- Gives you Grad-CAM visual explanations for transparency.
- Runs a rule-based analysis to measure severity.
- Shares recommendations based on what it finds.
- Presents all results through the Flask web interface.
- Prepares a final analysis report you can review.

Non-Functional Requirements

Performance: It doesn't keep you waiting long—a single image is processed quickly.

Usability. The interface is straightforward and easy to use.

Reliability: It responds properly even if someone uploads a bad or unsupported image.

Scalability: The architecture's modular, so you can add more disease types and models as needed.

Maintainability: Code is organized—frontend, backend, models, and utilities each live in their own module.

Security: All images and data are kept secure.

Explainability: You don't just get a label: the system shows you visual explanations alongside predictions.

4.8 ADVANTAGES OF THE HARDWARE AND SOFTWARE DESIGN

SkinGuard AI's hardware and software setup is built for speed and simplicity. Here's what makes it stand out:

Cost-Effective: You don't need fancy equipment. The system runs on everyday computers and uses open-source software, so it's easy on your budget.

Modular Design: Everything is split into different parts — preprocessing, segmentation, classification, Grad-CAM, and recommendations. This makes updates and fixes a breeze.

Efficient Image Processing: BM3D, CLAHE and Python handle tasks like resizing and normalizing skin images fast and smoothly.

Deep Learning Support: With TensorFlow and Keras, the system powers up U-Net and EfficientNetB3 for accurate lesion segmentation and classification.

Explainable Results: Grad-CAM shows you exactly which parts of the image the AI focused on, so you're not left guessing.

User-Friendly Interface: Flask, HTML, CSS, and JavaScript come together to give you a straightforward web interface. It's easy for anyone to use.

Scalable Architecture: The modular setup means you can add new models and features down the line without any headaches.

Faster Development: Python delivers a huge library pool — from AI to web development — so building and improving the app happens fast.

Local Execution: Once you've set up the models and libraries, everything runs right on your computer. No need to stay online constantly.

Easy Maintenance: By keeping the frontend, backend, AI models, and utility functions separate, tracking bugs or planning updates is way simpler.

4.9 SUMMARY

Here's the gist of SkinGuard AI's setup: it's built for quick, straightforward skin lesion analysis. On the backend, it runs on Python and Flask. The frontend uses HTML, CSS, and JavaScript to keep things accessible. For dealing with images, the system leans on OpenCV and NumPy to handle preprocessing and check image quality.

When it comes to actually identifying and classifying lesions, the backbone consists of two models — U-Net takes care of segmenting the lesions, and EfficientNetB3

classifies them. To help users see why the model decided one way or another, Grad-CAM adds visual explanations right to the results. There's also a rule-based module that gives information about severity and follows up with recommendations based on the analysis.

This modular design isn't just for show — it makes updates and add-ons a breeze. Pulling together deep learning and explainable AI means the system can actually help users walk through analyzing skin images step by step. Just remember, SkinGuard AI is here as a research and screening tool, not to replace your dermatologist.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 SOURCE CODE

5.1.1 Main Flask Application app.py

```
from flask import Flask, redirect, url_for

from routes.home import home

from routes.image_quality import image_quality

from routes.preprocessing import preprocessing

from routes.segmentation import segmentation

from routes.classification import classification

from routes.gradcam import gradcam

from routes.severity import severity

from routes.recommendation import recommendation

from routes.report import report

from routes.doctor import doctor

from routes.patient import patient

app = Flask(__name__)

app.secret_key = "skinguard_ai_secret_key_2026"

app.register_blueprint(home)

app.register_blueprint(image_quality)
```

```
app.register_blueprint(preprocessing)
app.register_blueprint(segmentation)
app.register_blueprint(classification)
app.register_blueprint(gradcam)
app.register_blueprint(severity)
app.register_blueprint(recommendation)
app.register_blueprint(report)
app.register_blueprint(doctor)
app.register_blueprint(patient)
@app.route("/")
def index():
    return redirect(
        url_for("home.home_page")
    )
@app.errorhandler(404)
def page_not_found(error):
    return """
    <h1>404 - Page Not Found</h1>
    <p>The requested page does not exist.</p>
    """, 404
if __name__ == "__main__":
```

```
    app.run(
        debug=True,
        host="127.0.0.1",
        port=5000
    )
```

5.1.2 U-Net Lesion Segmentation

```
from utils.segmentation_utils import segment_image
SEGMENT_FOLDER = "static/segmented"
os.makedirs(
    SEGMENT_FOLDER,
    exist_ok=True
)
def segmentation_page():
    latest_file = session.get(
        "preprocessed_image"
    )
    if not latest_file:
        return redirect(
            url_for(
                "preprocessing.preprocessing_page"
            )
```

```
    )
latest_file = latest_file.replace(
    "\\", "/"
).lstrip("/")
try:
    segmented_path = segment_image(
        latest_file,
        SEGMENT_FOLDER
    )
except Exception as e:
    print(
        "Segmentation Error:",
        str(e)
    )
    return (
        "Segmentation Failed: "
        + str(e)
    )
if not segmented_path:
    return "Segmentation output was not generated."
segmented_path = segmented_path.replace(
    "\\", "/"
```

```
    ).lstrip("/")
    session["segmentation_image"] = segmented_path
    session["segmentation_mask"] = segmented_path
    session["segmentation_completed"] = True
    return render_template(
        "segmentation.html",
        original_image="/" + latest_file,
        segmented_image="/" + segmented_path,
        algorithm="U-Net",
        status="Segmentation Completed Successfully"
    )
```

5.1.3 EfficientNetB3 Classification

```
from classification.predict import predict
@classification.route(
    "/classification/predict",
    methods=["POST"]
)
def classify_image():
    image_path = session.get(
        "uploaded_image"
    )
```

```
if not image_path:
    return jsonify({
        "success": False,
        "message": "No uploaded image found."
    })
image_path = normalize_image_path(
    image_path
)
if not os.path.isfile(image_path):
    return jsonify({
        "success": False,
        "message": "Uploaded image could not be found."
    })
# EfficientNetB3 Prediction
try:
    result = predict(
        image_path
    )
except Exception as e:
    return jsonify({
        "success": False,
        "message": str(e)
```

```
    })
# Classification Results
disease = result.get(
    "disease",
    "Unknown"
)
confidence = float(
    result.get(
        "confidence",
        0
    )
)
class_index = result.get(
    "class_index",
    0
)
probabilities = result.get(
    "probabilities",
    {}
)
# Save classification results
session["disease"] = disease
```

```
    session["confidence"] = confidence
    session["class_index"] = class_index
    session["probabilities"] = probabilities
    return jsonify({
        "success": True,
        "disease": disease,
        "confidence": confidence,
        "class_index": class_index,
        "probabilities": probabilities
    })
```

5.1.4 Grad-CAM Explainability

```
import os
import cv2
import tensorflow as tf
from gradcam.heatmap import generate_gradcam
MODEL_PATH = os.path.join(
    "models",
    "efficientnet_model.keras"
)
_model = None
def get_model():
```

```
    global _model
    if _model is None:
        _model = tf.keras.models.load_model(
            MODEL_PATH,
            compile=False
        )
    return _model
def generate_explanation(
    image_path,
    class_index,
    output_directory
):
    os.makedirs(
        output_directory,
        exist_ok=True
    )
    model = get_model()
    (
        overlay,
        important_region,
        heatmap
    ) = generate_gradcam(
```

```
    model=model,
    image_path=image_path,
    class_index=int(
        class_index
    )
)
original_filename = os.path.basename(
    image_path
)
image_name = os.path.splitext(
    original_filename
)[0]
gradcam_output = os.path.join(
    output_directory,
    image_name + "_gradcam.jpg"
)
important_output = os.path.join(
    output_directory,
    image_name + "_important.jpg"
)
overlay_bgr = cv2.cvtColor(
    overlay,
```

```
    cv2.COLOR_RGB2BGR
)
important_bgr = cv2.cvtColor(
    important_region,
    cv2.COLOR_RGB2BGR
)
cv2.imwrite(
    gradcam_output,
    overlay_bgr
)
cv2.imwrite(
    important_output,
    important_bgr
)
return (
    gradcam_output,
    important_output
)
```

5.2 SYSTEM SCREENSHOTS

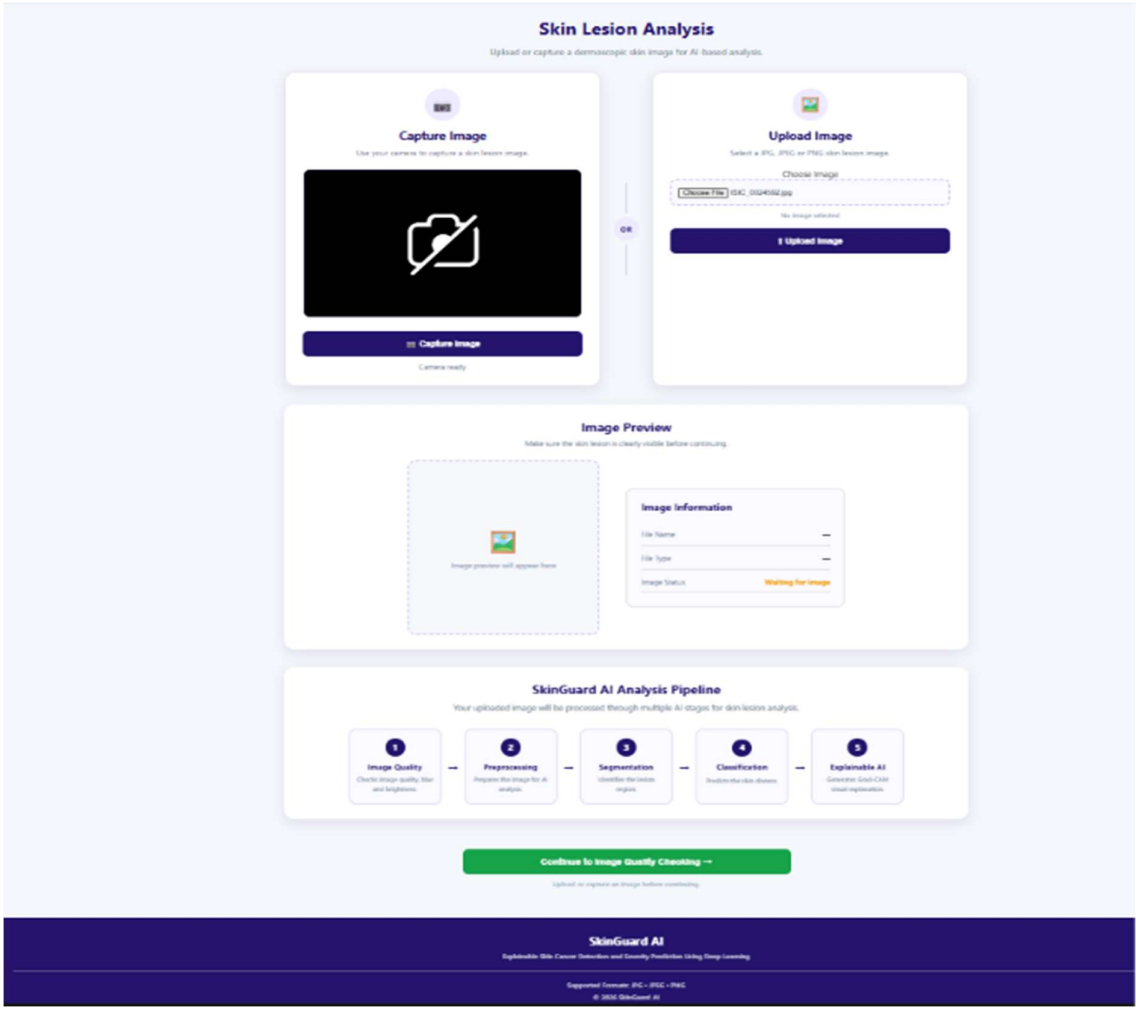


Figure 5.1 – SkinGuard AI Home / Upload Page

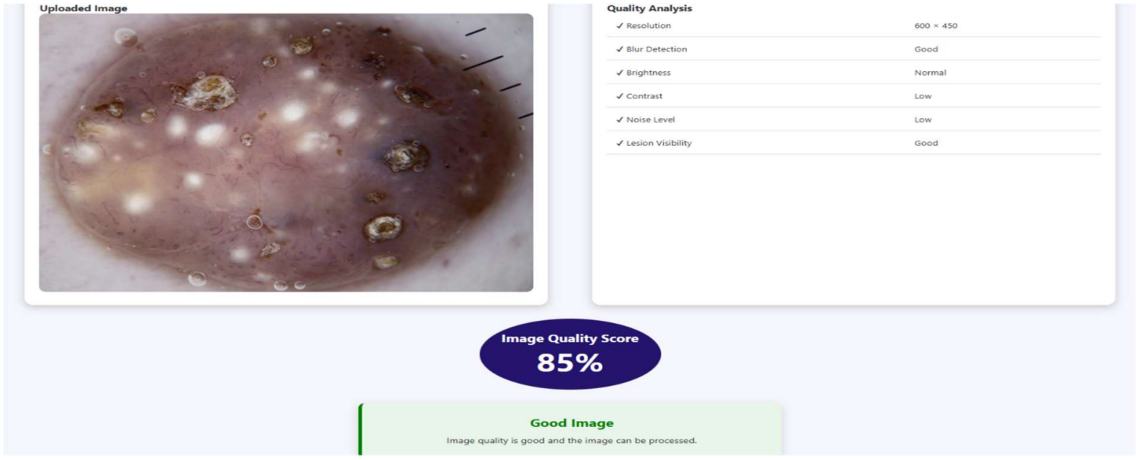


Figure 5.2 – Image Quality Check

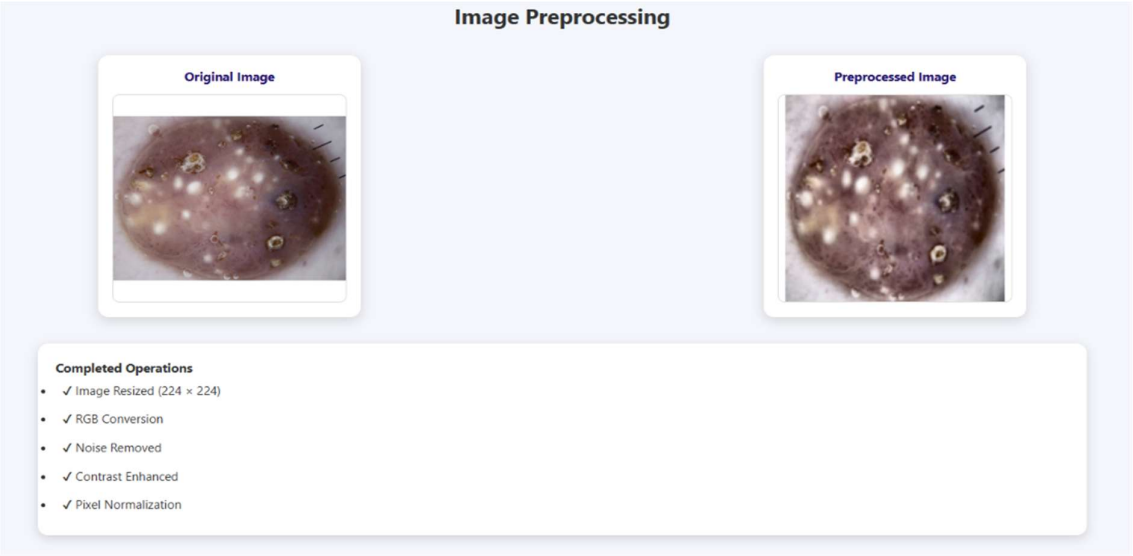


Figure 5.3 – Preprocessing Result

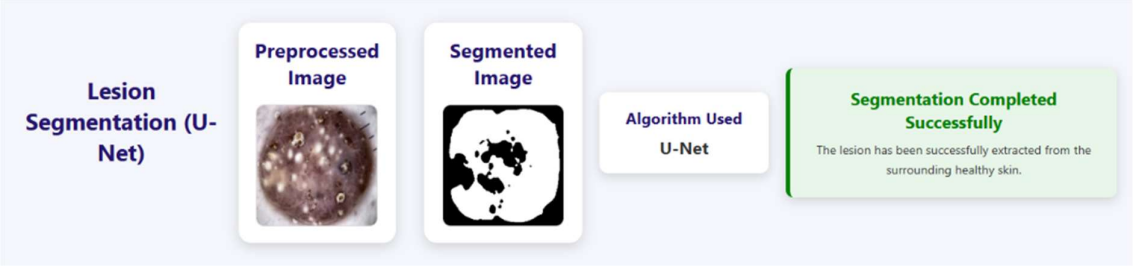


Figure 5.4 – U-Net Segmentation Result

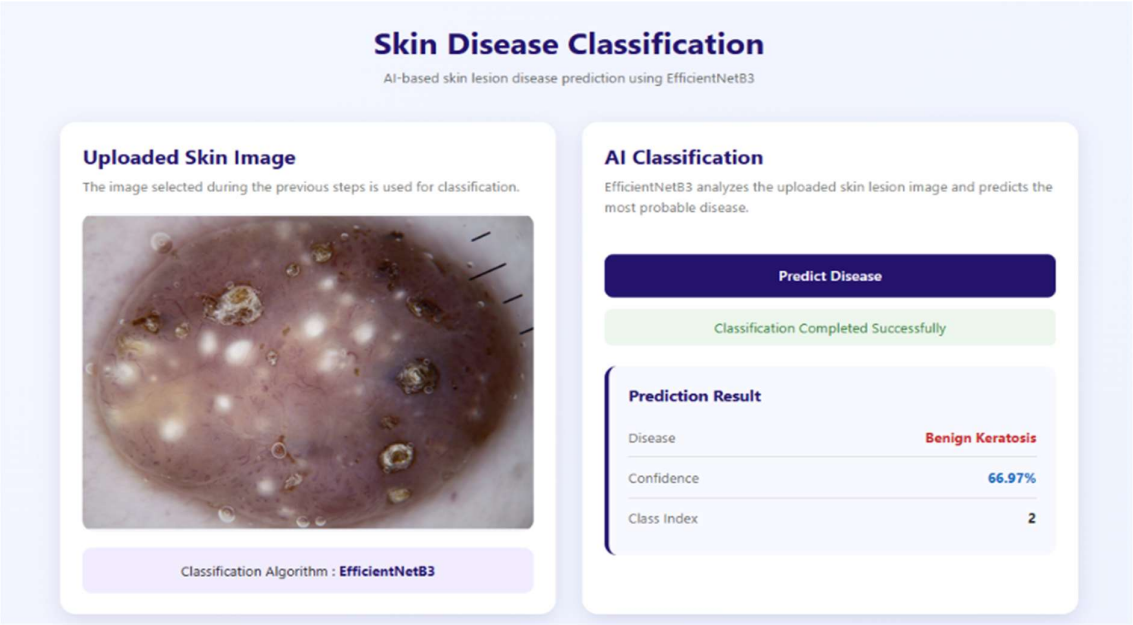


Figure 5.5 – EfficientNetB3 Classification Result

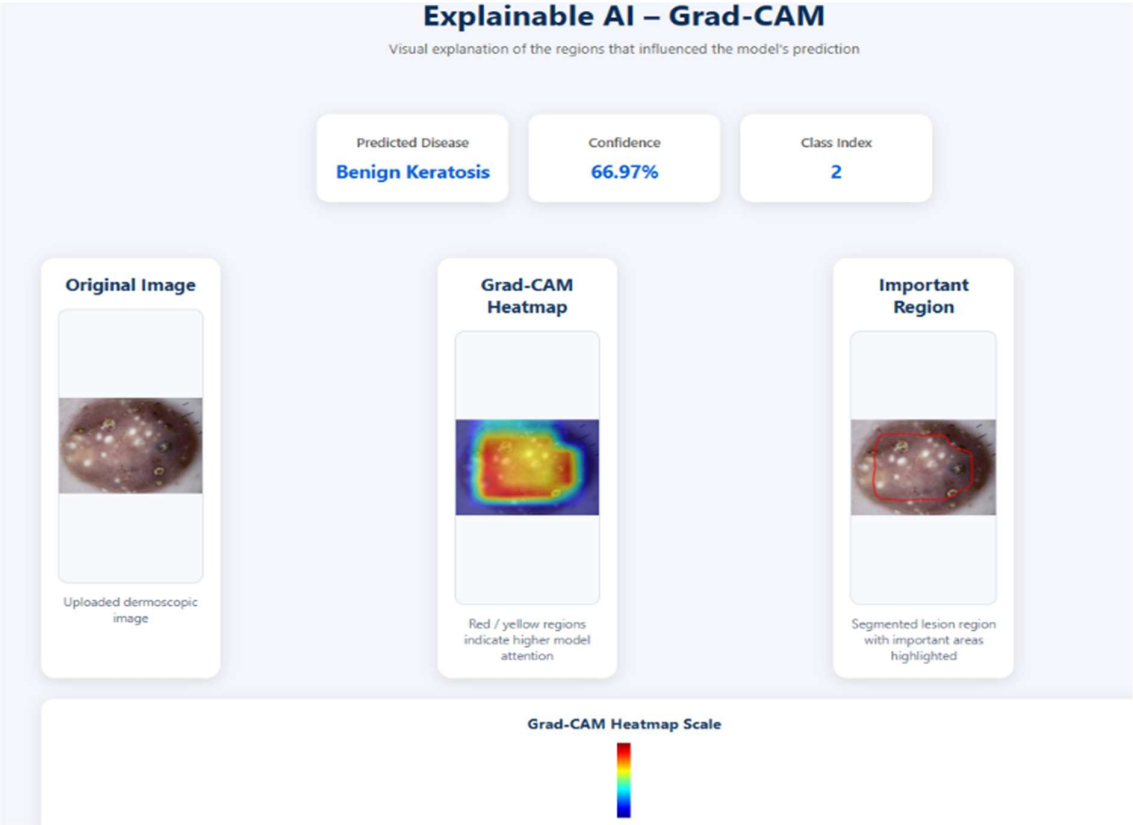


Figure 5.6 – Grad-CAM Explainability Result

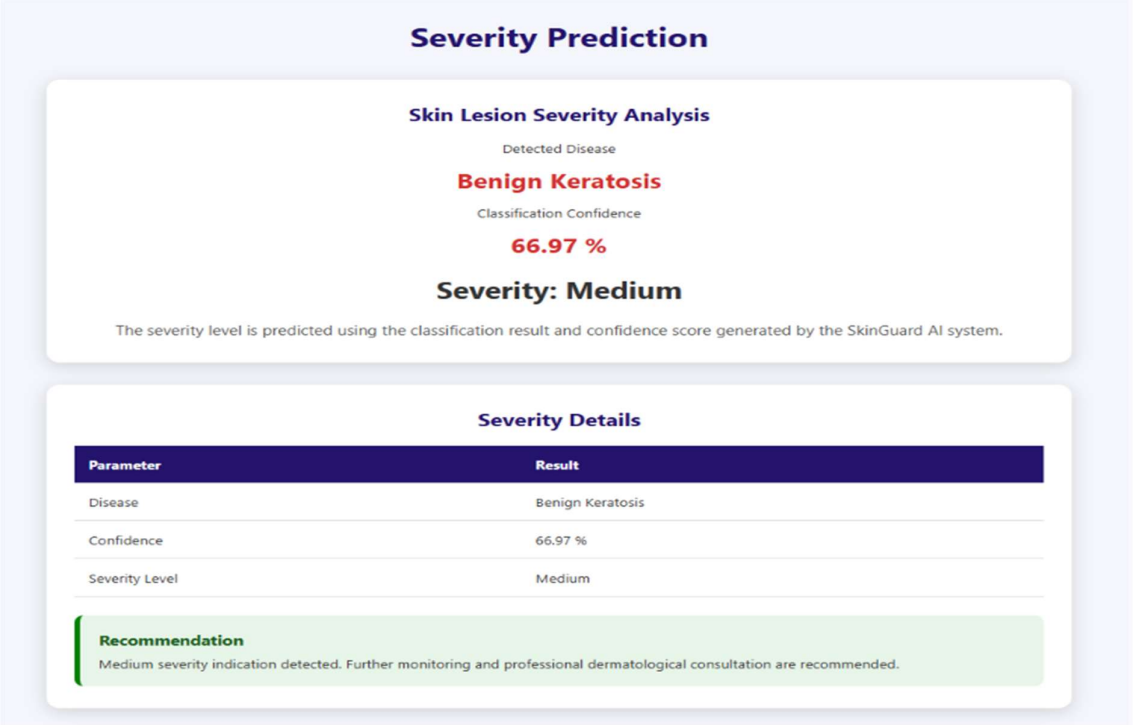


Figure 5.7 – Severity Prediction & Recommendation

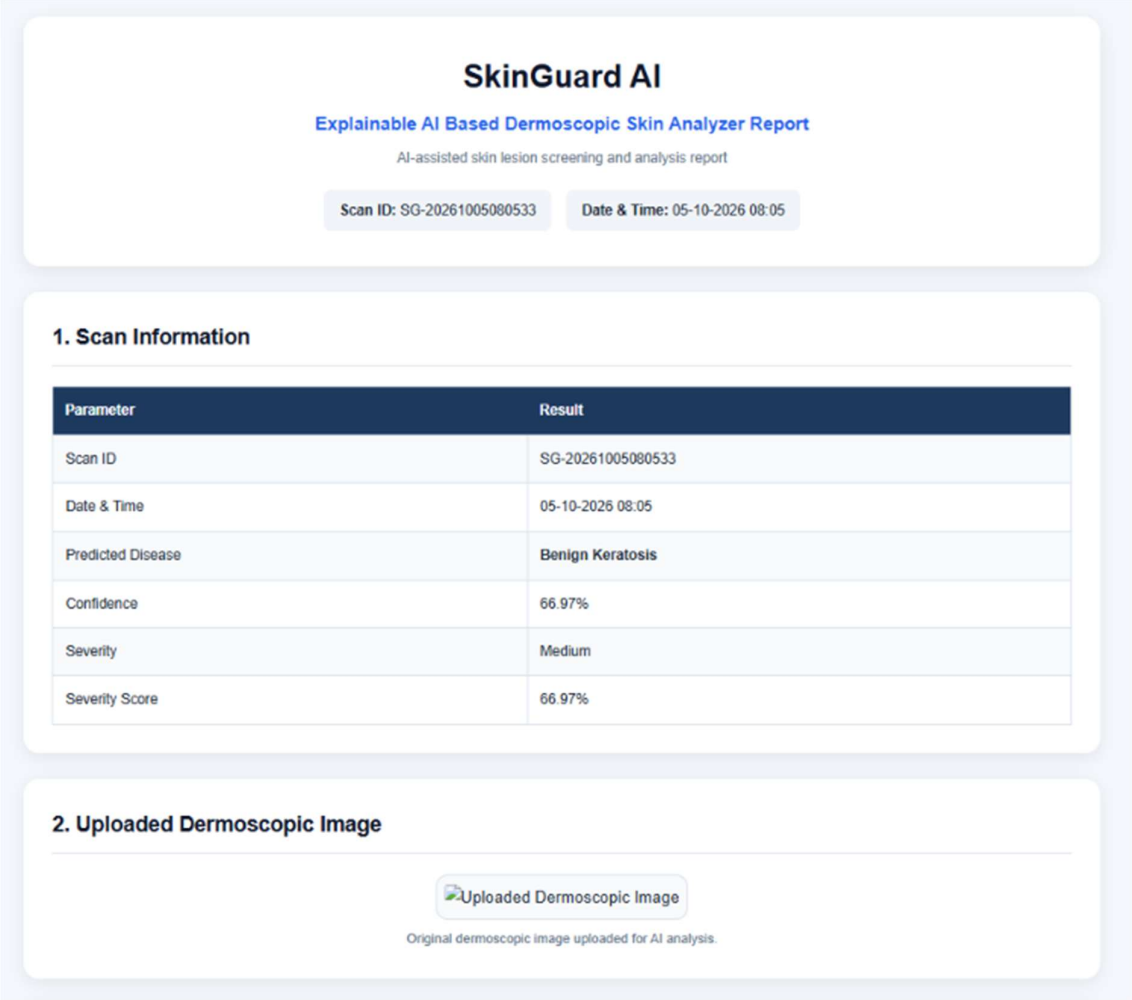


Figure 5.8 – Generated PDF Report

5.3 CLASSIFICATION RESULTS

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STARTING DISEASE PREDICTION

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Image Path : static/uploads/ISIC_0024592_1.jpg

Preprocessed Image Shape: (1, 224, 224, 3)

Pixel Minimum: 5.0

Pixel Maximum: 233.0

==========

ALL CLASS PROBABILITIES

==========

0 - Actinic Keratosis : 3.96%

1 - Basal Cell Carcinoma : 16.60%

2 - Benign Keratosis : 66.97%

3 - Dermatofibroma : 0.27%

4 - Melanoma : 6.13%

5 - Melanocytic Nevus : 5.83%

6 - Vascular Lesion : 0.24%

==========

TOP 3 PREDICTIONS

==========

1. Benign Keratosis : 66.97%

2. Basal Cell Carcinoma : 16.60%

3. Melanoma : 6.13%

==========

FINAL PREDICTION

==========

Class Index : 2

Disease : Benign Keratosis

Confidence : 66.97%

Probability Dictionary :

{'Actinic Keratosis': 3.96, 'Basal Cell Carcinoma': 16.6, 'Benign Keratosis': 66.97, 'Dermatofibroma': 0.27, 'Melanoma': 6.13, 'Melanocytic Nevus': 5.83, 'Vascular Lesion': 0.24}

==========

==========

CLASSIFICATION COMPLETED

==========

Image: static/uploads/ISIC_0024592_1.jpg

Disease: Benign Keratosis

Confidence: 66.97

Class Index: 2

Probabilities: {'Actinic Keratosis': 3.96, 'Basal Cell Carcinoma': 16.6, 'Benign Keratosis': 66.97, 'Dermatofibroma': 0.27, 'Melanoma': 6.13, 'Melanocytic Nevus': 5.83, 'Vascular Lesion': 0.24}

==========

5.4 GRAD-CAM EXPLAINABILITY RESULTS

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GRAD-CAM

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Original image: C:\Users\Varsha N\SkinGuard_AI\static/uploads/ISIC_0024592_1.jpg

==========

COMPLETE GRAD-CAM

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==========

GENERATING RAW GRAD-CAM

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Grad-CAM Input Shape: (1, 224, 224, 3)

EfficientNet Base Model Found: efficientnetb3

Grad-CAM Target Layer: top_activation

Classifier Layers:

- global_average_pooling

- feature_batch_normalization

- dropout_1

- dense_features

- dense_batch_normalization

- dropout_2

- classification_output

Prediction Shape: (1, 7)

Gradient Shape: (1, 7, 7, 1536)

Feature Map Shape: (1, 7, 7, 1536)

Raw Grad-CAM Heatmap: (7, 7)

Grad-CAM Generated Successfully

==========

Grad-CAM Legend Created: C:\Users\Varsha N\SkinGuard_AI\static\gradcam\gradcam_legend.jpg

Grad-CAM saved: static/gradcam/ISIC_0024592_1_gradcam.jpg

Important region saved: static/gradcam/ISIC_0024592_1_important.jpg

5.5 DISCUSSION

The developed SkinGuard AI system was tested using dermoscopic skin images. The system successfully followed the planned workflow from image preprocessing to final report generation.

The U-Net segmentation helped to identify the lesion region from the input image. The segmented output made the affected area easier to observe before classification.

The EfficientNetB3 model was used to classify the skin lesion into the required disease classes. The system displayed the predicted class along with its confidence value and class probabilities.

Grad-CAM was used to understand the prediction visually. The generated heatmap showed the important region of the image that contributed to the classification result. This makes the output easier to understand instead of showing only the predicted disease name.

Based on the classification result, the system provided a severity level and a suitable recommendation. The final PDF report combined the important results, including the input image, segmentation output, classification result, Grad-CAM result, severity, and recommendation.

Overall, the results show that the proposed system can provide a complete analysis in a single workflow. However, the system is intended as a supporting tool and not as a replacement for professional medical diagnosis.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

SkinGuard AI takes on skin lesion analysis with a smart, all-in-one web tool. It pulls together quality checks, image cleanup, U-Net segmentation, EfficientNetB3 classification, Grad-CAM for transparency, and rule-based severity scoring. You get everything in one place.

Here's how it works: U-Net spots and crops out the lesion from the original photo. EfficientNetB3 figures out what kind of lesion it is, tapping into what it's learned from tons of images. Grad-CAM then shows you exactly which parts of the image influenced its decision, making the process less of a mystery. On the user side, it's simple — you just upload or snap a picture using the Flask-based web interface, and the results pop up on the screen.

SkinGuard AI shows off what deep learning and explainable AI can do together. Plus, the system's flexible setup opens the door for future upgrades, like using more diverse data, improving accuracy, covering more skin issues, or adding smarter clinical tools.

6.2 FUTURE SCOPE

There's a ton of room to grow with SkinGuard AI. Here's how it could get even better down the line:

Bigger and Smarter Data

By training the system with more images—different skin tones, ages, and lighting conditions—it'll learn to handle just about anything you throw at it.

Sharper Models

Exploring newer CNNs and transformer-based approaches might give the AI a real boost in spotting and understanding skin issues.

Instant Analysis

Imagine snapping a photo with your phone or webcam and getting answers right away. Making real-time analysis possible could really change how people use it.

On-the-Go Apps

Building Android and iOS apps means anyone can check their skin concerns whenever they want, wherever they are. That's convenience.

More Explainers

Beyond Grad-CAM, weaving in other explainability tools would let users see exactly how and why the AI reaches its conclusions—both visually and in plain language.

Recognizing More

Expanding the classification module means the system won't just spot one or two skin problems, but a whole range of conditions.

Cloud Support

Putting the app on the cloud unlocks remote access for users and scalable processing for developers—making the tool more powerful and flexible.

Keep Getting Smarter

Regular updates using trusted datasets and expert feedback help the system stay sharp and accurate as new skin data rolls in.

Linking Up with Healthcare

With proper validation and regulations in place, SkinGuard AI could plug right into medical records or decision support systems—streamlining the process for doctors and patients.

Talks to Everyone

Adding multilingual support makes the app welcoming to people from all backgrounds, helping more folks get the skin care insights they need.

So, SkinGuard AI isn't just stopping here; it's got a clear path to becoming smarter, faster, and more useful for everyone.

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